25901708 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCHTC100 - PLANT PROPAGATION

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **10** questions

Each question carries **1** marks

1. Name any four tools used in plant propagation. [K] / [CO1]
2. Write any two types of artifical method of plant propagation. [K] / [CO2]
3. Compare and contrast the functions of the transfer area and incubation area in a tissue culture lab. [U] / [CO2]
4. Discuss the role of organic supplements in a tissue culture medium. [U] / [CO2]
5. Define Ring budding. [K] / [CO3]
6. What is a phytotron? [U] / [CO4]
7. Why are artificial propagation methods used in agriculture? [A] / [CO3]
8. What is callus culture? [K] / [CO3]
9. What is the difference between stem tuber and root tuber? [K] / [CO2]

10. How do auxins and cytokinins influence organogenesis in culture media? [U] / [CO2]

11. What is a rhizome? [K] / [CO3]

12. What is the role of Nitrogen in rooting? [U] / [CO3]

[1x10 = 10]

Part B

Short Essay

Answer any **6** questions

Each question carries **5** marks

13. Explain the role of sucrose as a carbon source in micropropagation media. [U] / [CO2]

14. Describe Veneer Grafting. [K] / [CO3]

15. Describe the basic design layout of a micropropagation laboratory. [U] / [CO2]

16. What are the advantages of vegetative propagation? [K] / [CO1]

17. What are the major challenges for the construction of a glass house? [A] / [CO4]

18. Explain bulbs, pseudobulbs, corms and tubers with examples. [U] / [CO3]

19. What are the applications of humidifiers in plant propagation? [U] / [CO4]

20. What is the basic principle behind layering? [K] / [CO3]

[5x6 = 30]

Part C

Essay

Answer any **1** questions

Each question carries **10** marks

21. Describe the unorganized methods of plant micropropagation and explain their significance in plant tissue culture [U] / [CO3]

22. Discuss the role of plant growth regulators in rooting. [U] / [CO3]

[10x1 = 10]